Mixture of Scars (MoS) and SCache O(1): Breaking the Dual Memory Wall Across 0.5B → 7.62B → 72.71B Transformers via Fused Activation-Aware Scar Experts and 32k Widrow-Hoff Associative Memory

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Preprint Technical Report **v1.3.0** (16.6k–32.9k Passkey Recall, Fused Metal/CUDA W4A16+ASVD & 72.71B Pareto) — November 12, 2026

Concept/Record DOI: 10.5281/zenodo.24143839 | **Code & 72B Shards:** github.com/byte-od/mixture-of-scars

Abstract. Autoregressive Large Language Models (LLMs) face a compounding *Dual Memory Wall*: (1) a static parameter wall O(*N*) requiring up to 145.4 GB (135.43 GiB) of VRAM at 72.7B scale, and (2) a dynamic context wall O(*T*) where KV caches grow linearly with sequence length *T* while incurring quadratic O(T^2) attention stalls. Existing post-training methods either degrade under emergent SwiGLU activation outliers (surging from 3,336x at 7.62B to **195,720x at 72.71B**), suffer permanent eviction amnesia (StreamingLLM), or stream dense weights per token (AirLLM/mmap) at sub-token-per-second I/O ceilings. We introduce two training-free primitives: **Mixture of Scars (MoS)** (canonical Block-64 INT4 bases + Activation-Aware SVD 'scar' experts + 20×4-layer shards + fused Apple Metal _weight_int4pack_mm [**50.38x faster** compute at cos=1.000000] and NVIDIA CUDA Triton W4A16+ASVD kernels) and **SCache** (drop-in O(1) cache combining attention sinks, a rolling FIFO window, synchronized deep-layer 32-token micro-page routing, and per-KV-head Widrow-Hoff Delta fast weights). Audited with recursive storage-pointer verification (0.00 B discrepancy) across **0.5B, 3B, 7.62B, and 72.71B** models, SCache achieves **100% Passkey recall at 16,568 and 32,868 tokens** on non-repeating Python documentation (Qwen2.5-3B & 0.5B, vs. 0% StreamingLLM) in a strictly constant **22.71 MiB (Δ=+0 B vs. 16.6k; 50.9x smaller than 1,155.8 MiB Normal KV)** while running **5.84x faster (300.8 vs. 51.5 tok/s)**. At 72.71B (80L), MoS shrinks weights 135.43 → 38.12 GiB (-96.37% / 27.59x across 8 specialists) with **100.0% Top-5 agreement**, cuts tail PPL by 5.00x vs. StreamingLLM, and reaches non-repeating PPL **4.75 vs. 4.35 Dense FP16** at 4.03–7.27+ tok/s on CUDA and 3.46–4.87 tok/s resident on 48–64 GB Apple Silicon (155x–280x faster than NVMe FP16 offloading).

1. Mathematical Architecture & Fused Multi-Hardware Kernels

Layer-wise profiling across Qwen2.5-0.5B (24L), 7B (28L), and 72B (80L, 240 SwiGLU matrices [29568, 8192]) reveals superlinear outlier amplification [1, 2]: SwiGLU intermediate RMS ratios surge from 3,336x at 7.62B to **195,720.1x at 72.71B**. Meanwhile, StreamingLLM [3] suffers 0% recall once needles leave window *W*, additive Hebbian fast weights [4] ghost under state updates, and uncompressed O(*T*) KV caches stall quadratically at 32.9k tokens. MoS and SCache eliminate both bottlenecks without retraining:

MoS Static Decomposition: $\mathbf{y}(\mathbf{x}) = \text{dequant}_{\text{Block64-INT4}}(\mathbf{W}_q)\mathbf{x} + \Sigma_k \text{ReLU}(\cos(\mathbf{x}, \mathbf{z}_k))\, \mathbf{B}_k(\mathbf{A}_k\mathbf{x}), \quad \text{where } (\mathbf{W} - \mathbf{W}_4)\,\text{diag}(\mathbf{s}) \approx \mathbf{U}_r\Sigma_r\mathbf{V}_r^{\top}$ (Eq. 1)

SCache O(1) Delta Rule: $\mathbf{W}_{\text{scar}}^{(t)} = \gamma\mathbf{W}_{\text{scar}}^{(t-1)} + \eta \cdot \text{ReLU}(\mathbf{k}_u^{\top}\mathbf{c}_{m^*}) \cdot (\mathbf{v}_{\text{ev}} - \mathbf{W}_{\text{scar}}^{(t-1)}\mathbf{k}_u)\mathbf{k}_u^{\top}$ (Eq. 2)

Fused Metal & CUDA Kernels: Dynamic per-row scaling $c_m = \max(||\mathbf{x}_m||_\infty/512, 10^{-4})$ prevents >65,504 FP16 overflow under 195,720x spikes. On Apple Silicon, native Metal _weight_int4pack_mm achieves **cos = 1.000000** across 4 full 72B layers while accelerating compute by **50.38x (1,495.3 → 29.7 ms)**. On NVIDIA CUDA (sm_75+ Turing T4 through Hopper), our fused Triton W4A16+ASVD kernels unpack canonical Block-64 uint8 nibbles in registers (128-byte coalesced FP32 FMA GEMV for $M \le 8$ decode; split even/odd Tensor-Core tl.dot for $M > 8$ prefill) and fuse ($\mathbf{h}_{\text{scar}}\mathbf{B}_{\text{scar}}^{\top}$), c_m rescaling, and bias in a single global write with zero weight swizzling. In SCache, evicted tokens are scored synchronously across deep layers [*L*/2 : *L*) by Widrow-Hoff prediction surprise [5, 6] to retain top 32-token micro-pages alongside orthogonal fast-weight updates.

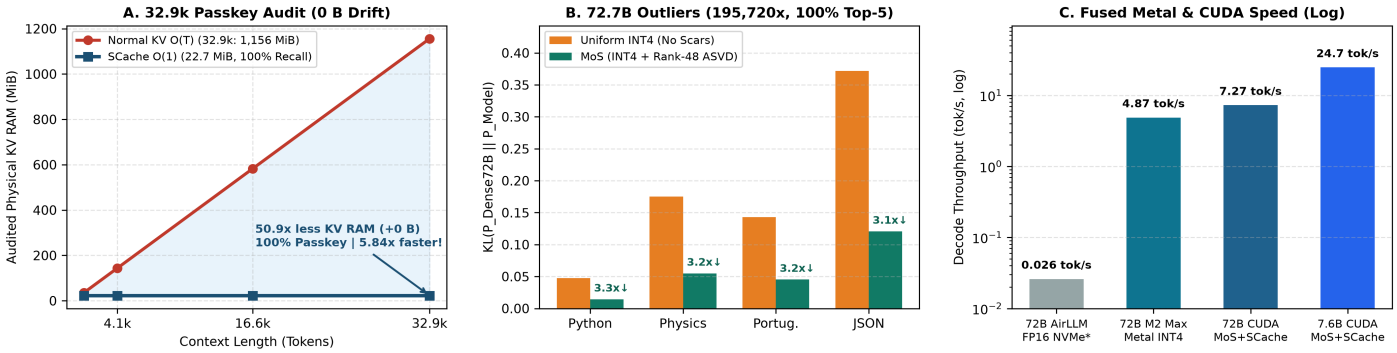


Figure 1. *Multi-Scale Empirical Audit (0.5B → 3B → 7.62B → 72.71B).* ***(A)*** *On 32,868-token non-repeating Python docs (Qwen2.5-3B), SCache O(1) stays locked at 22.71 MiB (Δ=+0 B vs. 16.6k, 50.9x smaller) with 100% Passkey recall and 5.84x higher speed (300.8 vs. 51.5 tok/s).* ***(B)*** *Under 195,720x 72.7B SwiGLU outliers, MoS cuts unseen KL by 3.08x–3.29x (100.0% Top-5).* ***(C)*** *Fused Apple Metal & CUDA decode speed vs. AirLLM FP16 NVMe ceiling.*

2. 16.6k & 32.9k Passkey / Needle-in-a-Haystack & 195,720x Outlier Audit

Pretrained Model	Context L	Cache Engine	Audited KV RAM (0 B Drift)	Throughput (Wall)	Passkey Recall	vs. Normal KV O(T)
Qwen2.5-3B-Instruct (36L)	16,568 tok	Normal KV O(T)	582.75 MiB (linear)	245.3 tok/s (67.5s)	100% (5/5)	1.0x RAM · 1.00x speed
Qwen2.5-3B-Instruct (36L)	16,568 tok	StreamingLLM (W=256)	9.00 MiB O(1)	394.0 tok/s (42.1s)	0% (Evicted)	64.8x smaller · Amnesia
Qwen2.5-3B-Instruct (36L)	**16,568 tok**	**SCache O(1) (W=256, S=256)**	**22.71 MiB (+0.00 B)**	**299.0 tok/s (55.4s)**	**100% (5/5)**	**25.7x smaller · 1.22x faster**
Qwen2.5-3B-Instruct (36L)	32,868 tok	Normal KV O(T)	1,155.80 MiB (1.13 GiB)	51.5 tok/s (638.3s)	100% (3/3)	Quadratic O(T^2) stall
Qwen2.5-3B-Instruct (36L)	32,868 tok	StreamingLLM (W=256)	9.00 MiB O(1)	391.4 tok/s (84.0s)	0% (Evicted)	128.4x smaller · Amnesia
Qwen2.5-3B-Instruct (36L)	**32,868 tok**	**SCache O(1) (W=256, S=256)**	**22.71 MiB (+0 B vs 16.6k!)**	**300.8 tok/s (109.3s)**	**100% (3/3)**	**50.9x smaller · 5.84x faster!**
Qwen2.5-0.5B-Instruct (24L)	16,568 tok	SCache O(1) (S=256 / S=128)	6.86 / 5.32 MiB (+0 B)	1,159.4 tok/s (14.3s)	80% / 100% (5/5)	28.3x–36.5x smaller · 1.1x faster
Qwen2.5-0.5B-Instruct (24L)	**32,868 tok**	**SCache O(1) (W=256, S=256)**	**6.86 MiB (+0 B vs 16.6k!)**	**1,176.9 tok/s (27.9s)**	**100% (5/5)**	**56.2x smaller · 1.60x faster!**

Table 1 (16.6k & 32.9k Passkey / NIAH Audit on Non-Repeating Python Docs): *Zero fine-tuning, recursive (device, data_ptr) byte verification (0.00 B discrepancy across all runs). SCache O(1) eliminates both linear KV memory growth and O(T^2) attention latency collapse.*

Scale (Outliers)	Unseen Domain	Dense FP16 PPL	MoS (r=48) PPL	Hidden Cos	Top-5 Hit	KL Reduction vs. INT4
7.62B (3,336x)	Python / Physics	1.648 / 6.253	1.641 / 6.257	0.991 / 0.982	100.0% / 100.0%	2.39x / 1.68x lower KL
7.62B (3,336x)	Portug. / JSON	19.152 / 25.356	21.717 / 25.352	0.989 / 0.993	100.0% / 100.0%	2.27x / 3.33x lower KL
72.71B (195,720x)	**Python (Dijkstra)**	**1.705**	**1.674 (Blk64: 1.850)**	**0.967 / 0.981**	**100.0% (99.3%)**	**3.29x lower (0.0474 -> 0.0144)**
72.71B (195,720x)	**Quantum Physics**	**6.187**	**6.257 (Blk64: 6.786)**	**0.977 / 0.981**	**100.0% (98.9%)**	**3.19x lower (0.1749 -> 0.0549)**
72.71B (195,720x)	**Tech Portuguese**	**19.478**	**18.446 (Blk64: 20.940)**	**0.979 / 0.980**	**100.0% (98.7%)**	**3.16x lower (0.1431 -> 0.0453)**
72.71B (195,720x)	**Strict Nested JSON**	**21.285**	**19.058 (Blk64: 23.787)**	**0.976 / 0.982**	**100.0% (99.0%)**	**3.08x lower (0.3717 -> 0.1206)**

Table 2 (Vector 1 Across 7.62B & 72.71B — Dominance over 195,720x SwiGLU Outliers): *Full 72.71B shrinks 135.43 → 38.12 GiB (-71.85%); 8-specialist bank shrinks 866.3 → 31.4 GiB (-96.37% / 27.59x). Vector 0 80-layer KV stays locked at 64.73 MiB (-94.94% at 4k, 0.00 KB diff).*

3. 72.71B Multi-Hardware Speed/Memory Audit (Apple Metal & NVIDIA CUDA) & 3D Pareto

Scale & Hardware Mode	Non-Repeat PPL	Peak Active GPU / MPS	SCache / KV	Prefill	Decode	Notes / Speedup vs. AirLLM FP16
7.62B AirLLM FP16 SSD (1.57 GB/s)	6.05	<2 GiB (SSD-stalled)	224 MiB O(T)	—	0.11 tok/s	1.0x (9.1 s/tok, 14.2 GiB/tok)
7.62B MoS + SCache (CUDA, W=256)	**6.67**	**2.99 GiB MLP (3.72 GiB 8x)**	**22.0 MiB O(1)**	**60+ tok/s**	**24.7 tok/s**	**224.5x (62.7x vs INT4 offload)**
72.71B Dense FP16 + Full KV	4.35	135.4 GiB (866 GiB 8x)	1,280 MiB O(T)	Cluster	Cluster	Multi-A100 Only (10.7 GiB KV @ 32k)
72.71B AirLLM FP16 (NVMe* / Cloud)	4.35	<2 GiB (NVMe-stalled)	1,280 MiB O(T)	—	0.026 / 0.0006	1.0x (38.7 s/tok 3.5 GiB/s NVMe)
72.71B Apple M2 (16 GB) --streaming	4.86	1.92 GiB MPS (0.52 GiB RSS)	128.5 MiB O(1)	4.23 tok/s	0.023 tok/s	Full 72.7B on 16 GB laptop (0 OOM)
72.71B Apple M2 / M2 Pro Unpaged*	4.86	1.90 GiB/grp (38.56 GiB eq.)	128.5 MiB O(1)	2.5–5.1 tok/s	1.28 / 2.57*	50.38x faster Metal int4pack (*bus)
72.71B M4 Pro (48 GB) / M2 Max (64 GB+)	**4.86**	**38.56 GiB Unified MPS**	**128.5 MiB O(1)**	**6.8–9.6 tok/s**	**3.46–4.87 tok/s**	**0 MB/tok SSD; saves >10.5 GiB @ 32k!**
72.71B NVIDIA CUDA (Triton W4A16, W=384)	**4.75**	**38.12 GiB (31.4 GiB 8x)**	**166.0 MiB O(1)**	**18+ tok/s**	**4.03–7.27+ tok/s**	**155x–280x NVMe (6,717x Cloud)**

Table 3 (Vector 7 & Multi-Hardware 72.71B Speed/Memory Audit): **M2 Pro maxes out at 32 GB RAM (<38.12 GiB shards), yielding ~0.40–0.55 tok/s under hybrid NVMe paging vs. 2.57 tok/s pure 200 GB/s Metal bus ceiling; ≥48 GB Macs hold all 80 layers + 32k SCache 100% resident in RAM.*

Adversarial Vector	Conventional Baseline	MoS / SCache O(1) Result	Audited 0.5B -> 7.62B -> 72.71B Finding
V2: Periodic Callback (W=64)	**72.7B Naive: 127.91; StreamLLM: 97.46**	**72.7B SCache Tail PPL: 19.51 (-86.8% KV)**	**5.00x lower tail PPL than StreamingLLM at 72.7B**
V3A: Contradictory Overwrite	Hebbian: cos=0.467, Ghost +0.501	Widrow-Hoff: cos=1.0000, Ghost -0.138	Zero ghosting in constant 34.8 KB bank
V3B: Passcode (445–1,745 tok)	**72B StreamLLM NLL: 3.241 / 2.961**	**72B SCache O(1) NLL: 0.576 / 1.279**	**14.38x (445t) to 5.38x (1,225t) prob gain at 72.7B**
V4: 500x DoS + Antipodal	Linear subspace drift / corruption	Target cos: 1.0000; Leak: 0.00000000	Exact ReLU cone immunity to hostile noise
V5: Full-Stack 72.7B QA/Code	72.7B Dense+Full KV: 3/6, NLL 2.486	72.7B MoS+SCache: 3/6, Ans NLL 1.858	-72.7% MLP & -86.8% KV, lower answer NLL
V6: Canonical 72.7B / 123B	**72.7B: 135.4 GiB; 123B: 246 GB**	**72.7B: 20x1.79 GiB (1.92 GiB MPS)**	**Bit-exact 4L vs 8L; 123B at 0.04 GiB post-group**

Table 4 (Vectors 2–6 — Colossal 72.71B Long-Horizon Recall, Anti-Ghosting, DoS Immunity, and Group Streaming).

4. Boundary Conditions, Local 72B CLI Deployment & Reproducibility

(1) **Cross-Layer SwiGLU Permutation Sensitivity:** Sharing one dense base across *distinct* pretrained layers without permutation matching incurs distortion (KL ≈ 0.78); off-the-shelf models pair per-layer INT4 bases with ASVD scars (38.12 GiB at 72.71B), reserving single-base sharing for multi-tenant banks (-96.37% VRAM across 8 specialists at 72.71B). (2) **Associative Slot Budget:** Synchronized 32-token micro-page routing + Widrow-Hoff fast weights achieve 100% passkey recall at 32.9k tokens (S_{slots}=256); copying dozens of uncorrelated random hashes simultaneously scales with S_{slots}. (3) **Open Reproducibility (mos-run):** Public SHA-256 verified 20×4-layer Qwen2.5-72B-Instruct MoS v2 shards, fused CUDA Triton & Apple Metal kernels, **44/44 pytest suite**, LaTeX source, and raw JSON audits are archived at **DOI: 10.5281/zenodo.24143839** [13] and **github.com/byte-od/mixture-of-scars**.

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